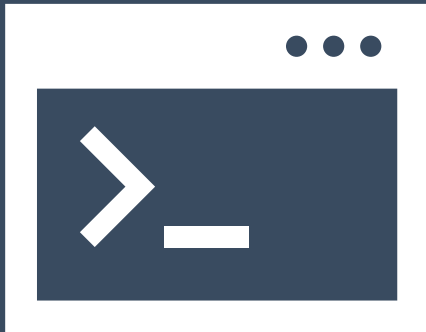


Introduction to Single-cell RNA-seq Analysis

<https://tinyurl.com/hbc-scrnaseq-online>



Harvard Chan Bioinformatics Core



Introductions!





Shannan Ho Sui
Director



Lorena Pantano
*Director of Bioinformatics
Platform*



John Quackenbush
Faculty Advisor



Upen Bhattarai



Will Gammerdinger



Noor Sohail



Alex Bartlett



Elizabeth
Partan



Emma Berdan



Zhu Zhuo



James Billingsley



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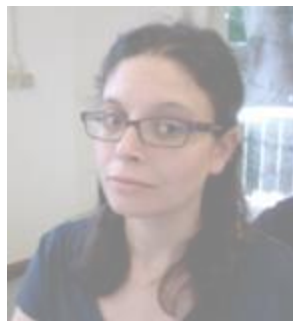
Noor Sohail



Alex Bartlett



Elizabeth
Partan



Emma Berdan



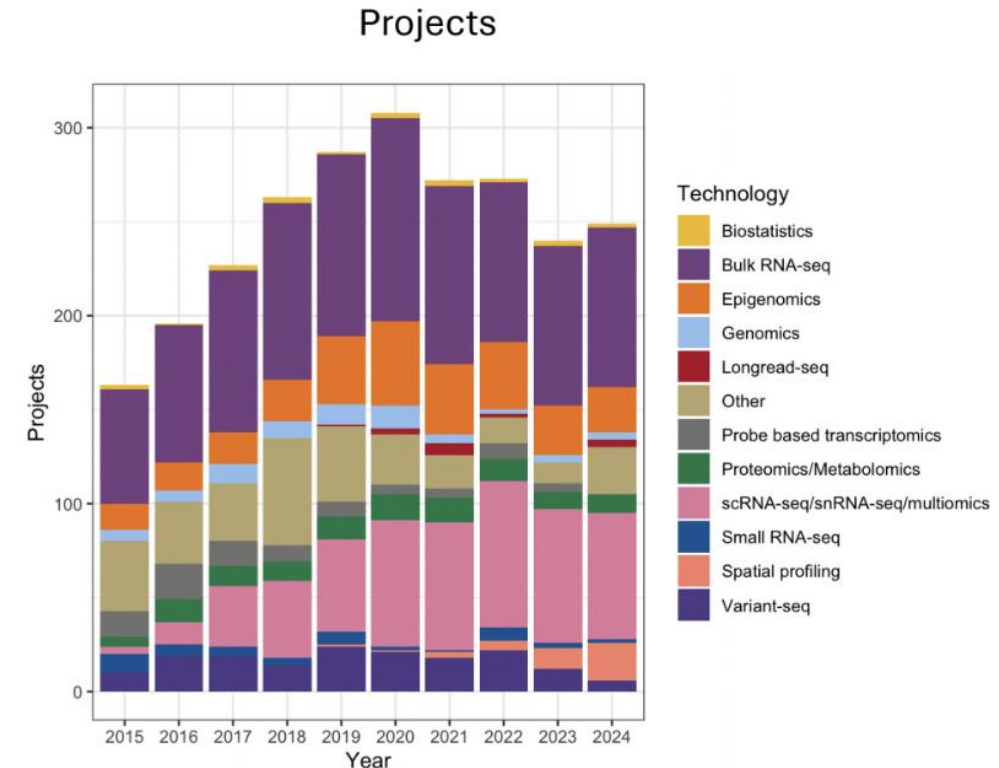
Zhu Zhuo



James Billingsley

Consulting

- ❖ Transcriptomics: Bulk, single cell, small RNA
- ❖ Epigenomics: ChIP-seq, CUT&RUN, ATAC-seq, DNA methylation
- ❖ Variant discovery: WGS, resequencing, exome-seq and CNV
- ❖ Multiomics integration
- ❖ Spatial biology
- ❖ Experimental design and grant support



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THE HARVARD CLINICAL
AND TRANSLATIONAL
SCIENCE CENTER



HARVARD
MEDICAL SCHOOL

Training

- ❖ Hands-on workshops design to reflect best practices, reproducibility and an emphasis on experimental design

- ❖ Basic Data Skills

- ❖ Shell

- ❖ R

- ❖ Advanced Topics: Analysis of high-throughput sequencing data

- ❖ Chromatin Biology & Peak Analysis

- ❖ Bulk RNA-seq

- ❖ Differential Gene Expression (Bulk & scRNA)

- ❖ scRNA-seq

- ❖ Variant Calling

- ❖ Current Topics in Bioinformatics

<https://bioinformatics.sph.harvard.edu/training>

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HARVARD
T.H. CHAN

SCHOOL OF PUBLIC HEALTH

DF/HCC

DANA-FARBER / HARVARD CANCER CENTER



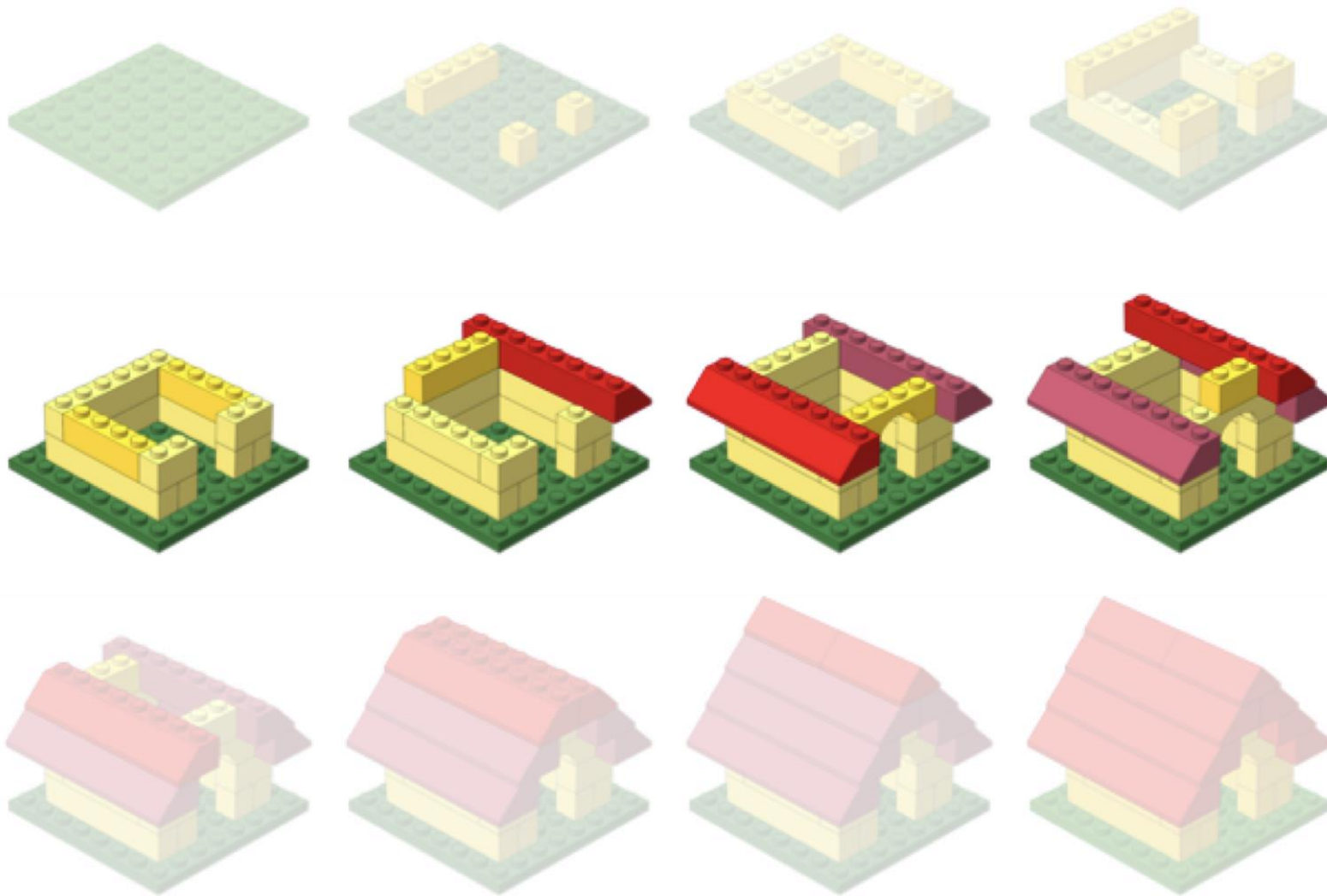
HARVARD
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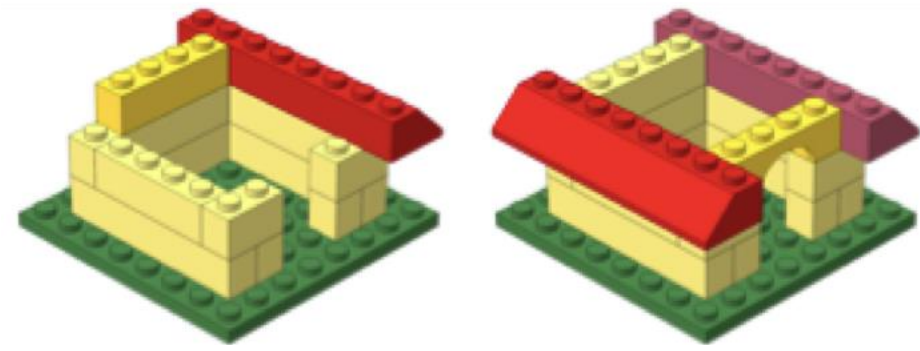
HARVARD
MEDICAL SCHOOL

Workshop scope



Bioinformatic Data Analysis

Workshop Scope



- ❖ Describe best practices for designing a single-cell RNA-seq experiment
- ❖ Describe steps in a single-cell RNA-seq analysis workflow
- ❖ Use Seurat and associated tools to perform analysis of single-cell expression data, including data filtering, QC, integration, clustering, and marker identification
- ❖ Understand practical considerations for performing scRNA-seq, rather than in-depth exploration of algorithm theory

Logistics



Course schedule

Day 1

Time	Topic	Instructor
09:30 - 09:45	<u>Workshop introduction</u>	Will
09:45 - 10:35	<u>Introduction to Single Cell RNA-sequencing: a practical guide</u>	<u>Dr. Mandovi Chatterjee</u>
10:35 - 10:40	Break	
10:40 - 11:00	scRNA-seq pre-reading discussion	All
11:00 - 11:45	<u>Quality control set-up</u>	Noor
11:45 - 12:00	Overview of self-learning materials and homework submission	Will

<https://tinyurl.com/hbc-scrnaseq-online>

Course materials

- ❖ We continuously update our materials to reflect changes in the field/software

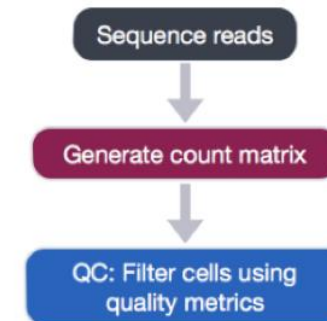


Approximate time: 90 minutes

Learning Objectives:

- Understand how to bring in data from single-cell RNA-seq experiments
- Construct QC metrics and associated plots to visually explore the quality of the data
- Evaluate the QC metrics and set filters to remove low quality cells

Single-cell RNA-seq: Quality control



<https://tinyurl.com/hbc-scrnaseq-online>

Single Screen & 3 Windows

The image shows a web browser window with a video conference and an RStudio interface. The video conference is titled "You are viewing Jihe Liu's screen" and shows three participants: Mary Piper, Troubleshooter (Radhika), and Jihe Liu. The RStudio interface is titled "Introduction To R / Intro-to-R" and shows a script editor with R code, a console with output, and a file explorer. The browser address bar shows "posit.cloud/spaces/657577/content/10438848".

Participants (3)

- Mary Piper (Co-host, me)
- Jihe Liu (Host)
- Troubleshooter (Radhika) (Co-host)

On this page

- [Code](#)
- [Learning Objectives](#)

Introduction To R / Intro-to-R

File Edit Code View Plots Session Build Debug Profile Tools Help

Intro-to-R.R

```
1 3 + 5
2
3 # Intro to R Lesson
4 # June 3rd, 2025
5
6 # Interacting with R
7
8 ## I am adding 3 and 5. R is fun!
9
10 # Top Level
```

Console

```
Type 'demo()' for some demos, 'help()' for on-line help, or
'help.start()' for an HTML browser interface to help.
Type 'q()' to quit R.

> 3 + 5
[1] 8
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>
```

Files

- ..
- .Rhistory
- data
- project.Rproj
- scripts
- results
- figures

Environment

Environment is empty

Files

Name	Size	Modified
..		
.Rhistory	0 B	May 28, 2025, 4:43 PM
data		
project.Rproj	205 B	May 30, 2025, 2:51 PM
scripts		
results		
figures		

*Our
Recommendation*

Single Screen & 3 Windows

Zoom

Our Recommendation

Participants (3)

- Mary Piper (Co-host, me)
- Jihe Liu (Host)
- Troubleshooter (Radhika) (Co-host)

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On this page

[Code](#) [Learning Objectives](#)

Single Screen & 3 Windows

You are viewing Jihe Liu's screen

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HBC GitHub Contact us

Web Browser

Code Learning Objectives

Introduction To R / Intro-to-R

posit.cloud/spaces/657577/content/10438848

File Edit Code View Plots Session Build Debug Profile Tools Help

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Environment

R - Global Environment

Environment is empty

Files Plots Packages Help Viewer Presentation

- .Rhistory
- data
- project.Rproj
- scripts
- results
- figures

Our Recommendation

Single Screen & 3 Windows

The image displays three overlapping windows from a single screen:

- Video Conference Window:** Shows a meeting with three participants: Mary Piper, Troubleshooter (Radhika), and Jihe Liu. The interface includes a 'View Options' dropdown and a 'Participants (3)' list.
- Web Browser Window:** Displays the posit.cloud website. The address bar shows 'posit.cloud/spaces/657577/content/10438848'. The page has a navigation bar with links for HBC, GitHub, and Contact us. Below the navigation bar, there is a section titled 'On this page' with a link for 'Learning Objectives'.
- RStudio Window:** Shows the RStudio IDE interface. The title bar indicates 'Intro-to-R - Posit Cloud'. The editor pane contains R code for an 'Intro to R Lesson' dated June 3rd, 2025. The console pane shows the output of the code, including the result of a calculation (3 + 5 = 8). The file explorer pane shows the project structure, including folders for .Rhistory, data, project.Rproj, scripts, results, and figures.

Our Recommendation

RStudio

Single Screen & 3 Windows

Zoom

Web Browser

RStudio

Our Recommendation

Participants (3)

- Mary Piper (Co-host, me)
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Web Browser

Code

Learning Objectives

Introduction To R / Intro-to-R

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Environment

History

Connections

Tutorial

Environment is empty

Files

Plots

Packages

Help

Viewer

Presentation

Cloud

project

0 B May 28, 2025, 4:43 PM

205 B May 30, 2025, 2:51 PM

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> 3 + 5
[1] 8
>
```

Course participation

- ❖ Mandatory review of self-learning lessons and assignments
- ❖ Attendance required for all classes
- ❖ Your questions and active participation drive learning
- ❖ **We look forward to all of your questions!**



Course participation

- ❖ At-home lessons and exercises after each session
- ❖ Cover material not previously discussed
- ❖ Provides us feedback to help pace the course appropriately
- ❖ 3-5 hours to complete

Using AI for Assignments

❖ Do

- ❖ Try to resolve error messages with it
- ❖ Test code written by AI on a dataset where you have expected results
- ❖ Take the time to review the generated code line-by-line

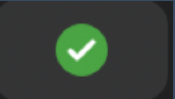
❖ Don't

- ❖ Implement it in replacement to learning
- ❖ Write code that you don't understand
- ❖ Assume the output from an AI process is correct

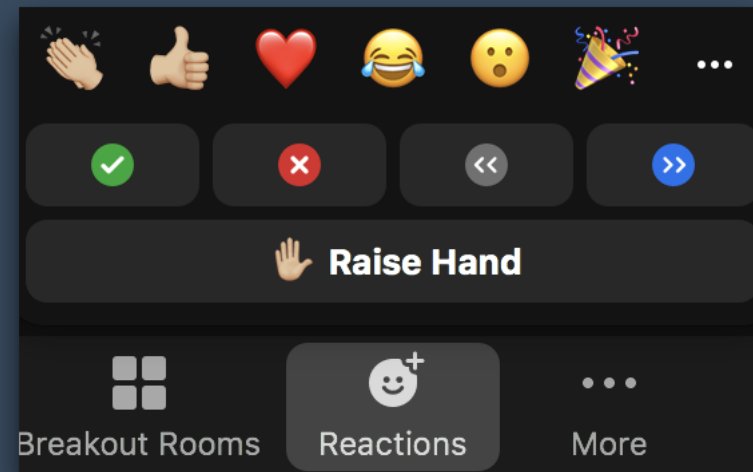
Odds & Ends

❖ Quit/minimize all applications that are not required for class

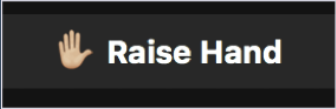
❖ Are you all set?

❖  = "agree", "I'm all set"

❖  = "disagree", "I need help"



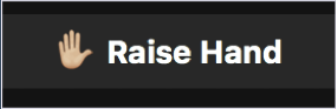
Odds & Ends

- ❖ Questions for the presenter?
 - ❖ Post the question in the Chat window OR
 - ❖  when the presenter asks for questions
 - ❖ Let the Moderator know

Odds & Ends

❖ Questions for the presenter?

- ❖ Post the question in the Chat window OR

- ❖  when the presenter asks for questions

- ❖ Let the Moderator know

❖ Technical difficulties with software?

- ❖ Start a private chat with the Moderator with a description of the problem

Contact Us

- ❖ *HBC training team:* hbctraining@hsph.harvard.edu
- ❖ *HBC consulting:* bioinformatics@hsph.harvard.edu